

Experiment – 07 | Database Normalization

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Topic: Database Normalization – Functional Dependencies, Candidate Keys and Normal Forms

Reference: The supplied Experiment 07 material covers functional dependency, closure, candidate keys, prime/non-prime attributes, 1NF, 2NF, 3NF and BCNF.

Question 1: Candidate Keys

Relation: $R(A, B, C, D)$
Functional Dependencies:
 $AB \rightarrow C$
 $C \rightarrow D$
 $D \rightarrow A$

Solution:
 $(AB)^+ = ABCD \rightarrow AB$ is a candidate key.
 $(BC)^+ = BC \rightarrow D \rightarrow A \rightarrow BC$ is a candidate key.
 $(BD)^+ = BD \rightarrow A \rightarrow C \rightarrow BD$ is a candidate key.

Therefore:
Candidate Keys = {AB, BC, BD}
Prime Attributes = {A, B, C, D}
Non-Prime Attributes = { }

Question 2: Candidate Keys

Relation: $R(A, B, C, D, E)$
Functional Dependencies:
 $A \rightarrow D$
 $B \rightarrow A$
 $BC \rightarrow D$
 $AC \rightarrow BE$

Solution:
 $(AC)^+ = AC \rightarrow D, B, E \rightarrow AC$ is a candidate key.
 $(BC)^+ = BC \rightarrow A \rightarrow D$ and $AC \rightarrow E \rightarrow BC$ is a candidate key.

Therefore:
Candidate Keys = {AC, BC}
Prime Attributes = {A, B, C}
Non-Prime Attributes = {D, E}

Question 3: Candidate Keys

Relation: $R(A, B, C, D, E)$
Functional Dependencies:
 $B \rightarrow A$
 $A \rightarrow C$
 $BC \rightarrow D$
 $AC \rightarrow BE$

Solution:

$A^+ = A \rightarrow C; AC \rightarrow B, E; BC \rightarrow D$

Therefore $A^+ = ABCDE$, so A is a candidate key.

$B^+ = B \rightarrow A \rightarrow C; BC \rightarrow D; AC \rightarrow E$

Therefore $B^+ = ABCDE$, so B is a candidate key.

Therefore:

Candidate Keys = {A, B}

Prime Attributes = {A, B}

Non-Prime Attributes = {C, D, E}

Question 4: Candidate Keys

Relation: R(A, B, C, D, E, F)

Functional Dependencies:

$A \rightarrow BCD$

$BC \rightarrow DE$

$B \rightarrow D$

$D \rightarrow A$

Solution:

$A^+ = ABCDE$, but F is not obtained. Therefore A is not a key by itself.

$AF^+ = ABCDEF \rightarrow AF$ is a candidate key.

$BF^+ = ABCDEF$ because $B \rightarrow D \rightarrow A$ and $A \rightarrow C$; therefore BF is a candidate key.

$DF^+ = ABCDEF$ because $D \rightarrow A \rightarrow B, C$ and $BC \rightarrow E$; therefore DF is a candidate key.

Therefore:

Candidate Keys = {AF, BF, DF}

Prime Attributes = {A, B, D, F}

Non-Prime Attributes = {C, E}

Practice Set – Highest Normal Form and Candidate Keys

Practice 1. $R(A,B,C,D,E)$, FDs: $CE \rightarrow D$, $D \rightarrow B$, $C \rightarrow A$

$C^+ = C,A$; $CE^+ = C,E,D,B,A = ABCDE$, so $CK = \{CE\}$. $C \rightarrow A$ is a partial dependency on the composite key CE ; hence not 2NF. Highest normal form: 1NF.

Practice 2. $R(A,B,C,D,E,F)$, FDs: $AB \rightarrow C$, $DC \rightarrow AE$, $E \rightarrow F$

$AB^+ = ABCDEF$, so $CK = \{AB\}$. $AB \rightarrow C$ is fine for 2NF. But $E \rightarrow F$ has a determinant that is not a super key and F is non-prime, so 3NF is violated. Highest normal form: 2NF.

Practice 3. $R(A,B,C,D,E,F,G,H,I)$, FDs: $AB \rightarrow C$, $BD \rightarrow EF$, $AD \rightarrow GH$, $A \rightarrow I$

$ABD^+ = ABCDEFGHI$, so $CK = \{ABD\}$. $A \rightarrow I$ is a partial dependency because A is a proper subset of ABD and I is non-prime. Highest normal form: 1NF.

Practice 4. $R(A,B,C,D,E)$, FDs: $AB \rightarrow CD$, $D \rightarrow A$, $BC \rightarrow DE$

$BC^+ = ABCDE$, so BC is a CK. $D^+ = DA$ only, and using A with B is needed for the full relation. Candidate keys include BC and BD . $D \rightarrow A$ has determinant D that is not a super key and A is prime, so 3NF can still hold; however $BC \rightarrow DE$ is a super-key dependency. Highest normal form: 3NF.

Practice 5. $R(A,B,C,D,E)$, FDs: $BC \rightarrow ADE$, $D \rightarrow B$

$BC^+ = ABCDE$, so BC is a CK. $D \rightarrow B$ has determinant not a super key and B is prime, so 3NF is satisfied. Since D is not a super key, BCNF is violated. Highest normal form: 3NF.

Practice 6. $R(A,B,C)$, FDs: $A \rightarrow B$, $B \rightarrow C$, $C \rightarrow A$

A^+ , B^+ , and C^+ each give ABC . Candidate keys are $\{A\}$, $\{B\}$, $\{C\}$. All attributes are prime. Every determinant is a candidate key, so the relation satisfies BCNF. Highest normal form: BCNF.

Additional Problem: Relation P

Fields: F_1, F_2, F_3, F_4, F_5 . FDs: $F_1 \rightarrow F_3$, $F_2 \rightarrow F_4$, $(F_1,F_2) \rightarrow F_5$.

Candidate key = $\{F_1,F_2\}$, because F_1F_2 determines all five attributes. $F_1 \rightarrow F_3$ and $F_2 \rightarrow F_4$ are partial dependencies on proper subsets of the composite key, with F_3 and F_4 non-prime. Therefore the relation is in 1NF but not 2NF. Prime attributes = $\{F_1,F_2\}$; Non-prime attributes = $\{F_3,F_4,F_5\}$.